



The 2011 Nuclear Phase-Out and Thermal Load

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Presentation of Thesis Topic | 18. Juni 2026

1. Motivation: cooling water as thermal pollution

The sudden **shutdown of eight nuclear power plants in 2011** is used to investigate whether the elimination of cooling-water demand **reduces river temperatures and preserves dissolved oxygen levels**.

THE SHOCK: 6. August 2011

After Fukushima, Germany abruptly and permanently shut down **eight older nuclear plants** in summer 2011 (incl. Biblis, Neckarwestheim I, Isar I) .



Cooling-water withdrawal

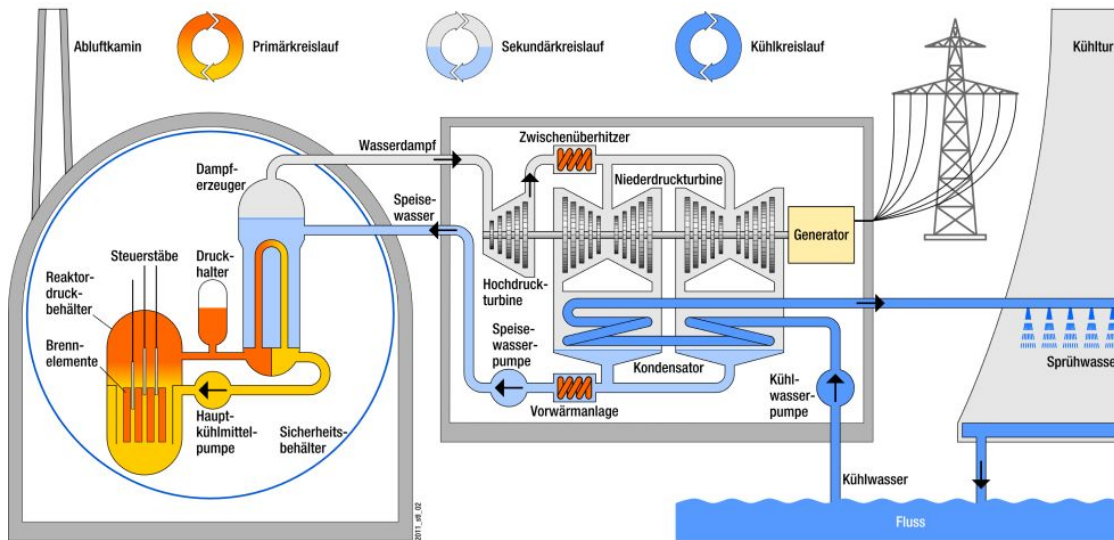


Heated water discharged



↓ Dissolved oxygen

2. Research question



„What is the causal effect of the abrupt 2011 shutdown of **nuclear power plants** on summer water **temperature** and **dissolved-oxygen** levels in German rivers?“

3. Identification strategy: Difference-in-Differences

TREATMENT GROUP

River monitoring stations immediately **downstream** of the 8 plants decommissioned in 2011.

e.g. Biblis · Neckarwestheim I · Isar I

CONTROL GROUP

Stations downstream of plants **exempt from the shutdown that stayed at full load** after 2011.

e.g. Isar II · Neckarwestheim II · Grohnde

SHOCK

Summer 2011

OUTCOMES

Summer water temp (°C) dissolved O₂ (mg/l)

4. Nuclear reactors map



Nuclear reactors shut down as part of the nuclear moratorium



Those still in operation after 2011

Before 2011 both groups move in parallel.
After 2011 temperature should fall further at treated stations once the heat source is removed.

5. Empirical specification: two-way fixed-effects DiD

$$Y_{s,t} = \beta (Treated_s \times Post2011_t) + \alpha_s + \delta_t + \theta W_{s,t} + \varepsilon_{s,t}$$

$Y_{s,t}$ = summer temp (°C) or dissolved O₂ (mg/l), station s , year t

$Treated_s$ = downstream of a 2011-decommissioned plant

$Post2011_t$ = 1 for years ≥ 2011

α_s, δ = station & year fixed effects (absorb hot-year shocks)

$W_{s,t}$ = regional weather controls (air temp, discharge ...)

β = causal estimate of interest

Expected: **temperature $\beta < 0$** dissolved O₂ $\beta > 0$ *extension: event-study to test pre-trends*

6. Core datasets: outcome & treatment



OUTCOME

EEA Waterbase: Water Quality ICM

USE

DisaggregatedData (dated raw samples) + *SpatialObject* (station coordinates)

KEY FIELDS

monitoringSiteIdentifier · countryCode=DE · water-body category · determinand (temp / O₂) · sampling date · value



TREATMENT

Conventional power plants (OPSD)

USE

conventional_power_plants_DE.csv

KEY FIELDS

energy_source = Nuclear · shutdown · status · lat / lon · commissioned

GROUPS

Treated = shutdown 2011 (8 plants) Control = nuclear still running after 2011 (e.g. Isar 2, Neckarwestheim 2, Grohnde)

7. Additional datasets: controls & geography



WEATHER

DWD Climate Data Center

Air temperature,
precipitation,
sunshine daily, per station.
Absorbs hot-summer variation.



DISCHARGE

GRDC · PEGELONLINE

Summer low flow concentrates the
thermal load a key control and
mechanism.



RIVER NETWORK

EU-Hydro (Copernicus)

Flow direction & topology defines which
stations are downstream; spatial join in R
(sf).



Linkage: plant coordinates → nearest *downstream* monitoring station via EU-Hydro; weather & discharge joined by nearest station / gauge.

8. Critical evaluation

STRENGTHS

- Clean, exogenous and **precisely dated shock** (the 2011 phase-out).
- Reliable, open data: EEA Waterbase monitoring.

LIMITATIONS

- Few **treated sites (8 plants)** → limited statistical power.
- Defining “**downstream**” and matching stations to plants is non-trivial.
- **River flow & seasonal dynamics** add noise.

9. Next steps

1

Confirm data & build the station plant map

Pull Waterbase, OPSD, DWD & discharge; assign downstream stations via EU-Hydro.

2

Construct the panel & run the DiD

Summer aggregates per station-year; two-way FE with weather controls.

3

Robustness

Event-study pre-trends, placebo timing, alternative controls, distance bands.

DATA STACK

EEA Waterbase outcome: temp + O₂

OPSD plants treatment / shutdown

DWD CDC weather controls

GRDC · PEGELONLINE river discharge

EU-Hydro river network